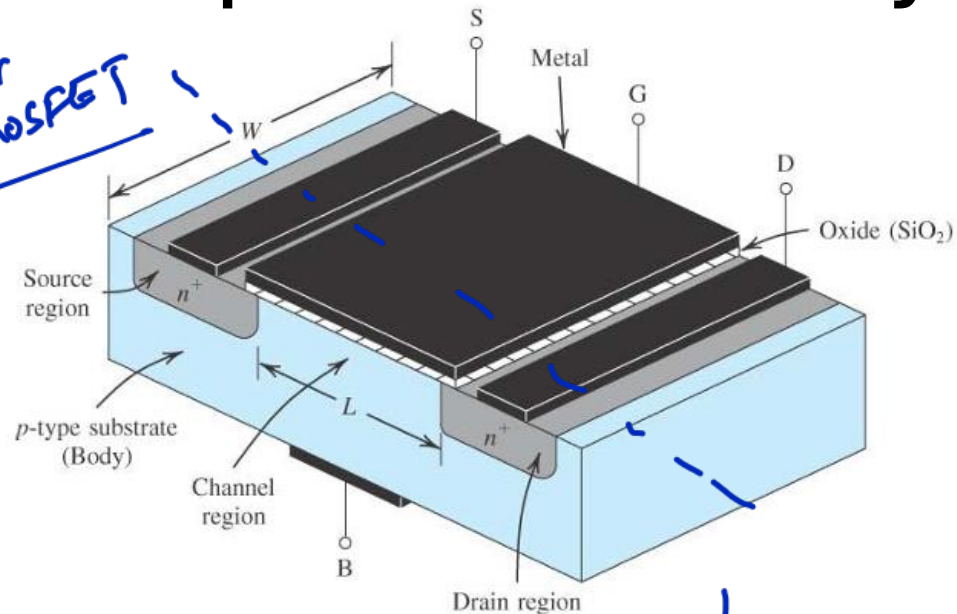
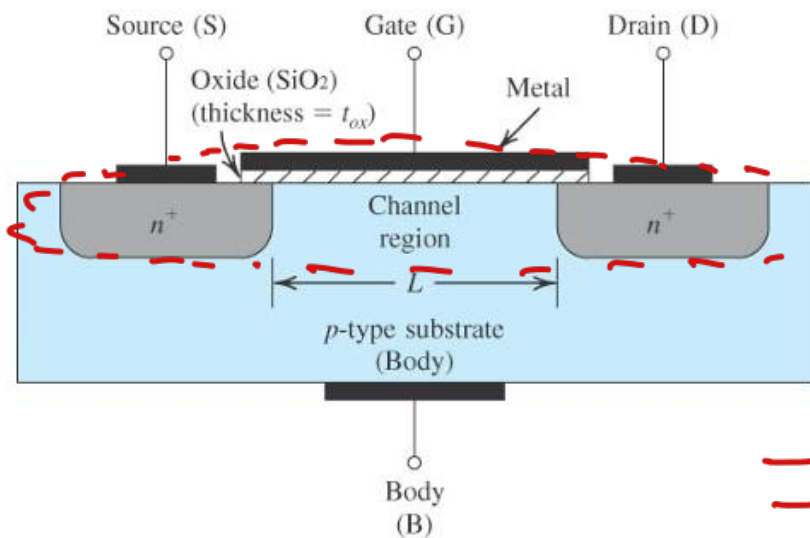


Recap: MOSFET Physics – Band Diagrams

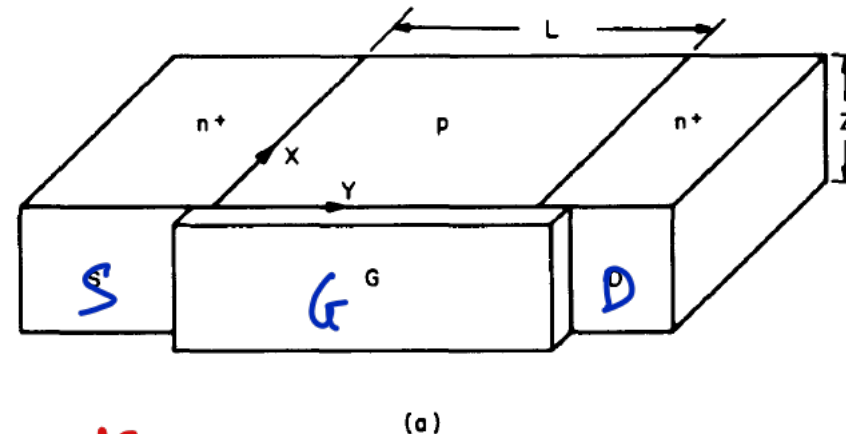
planar mosfet



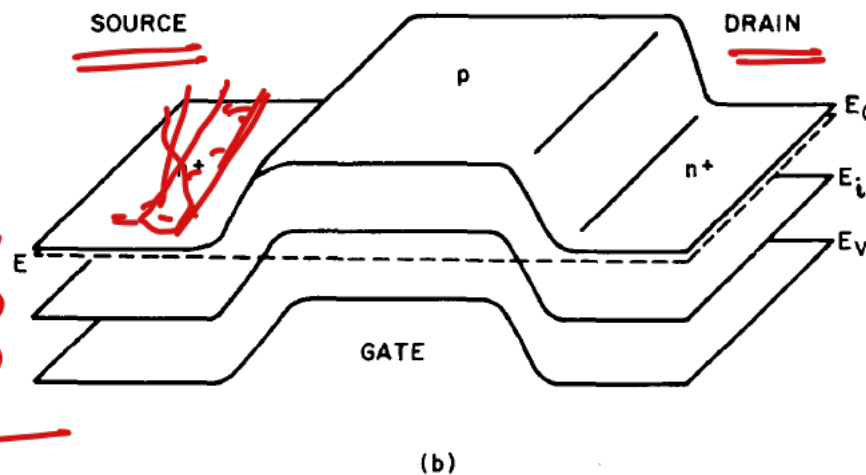
Front View



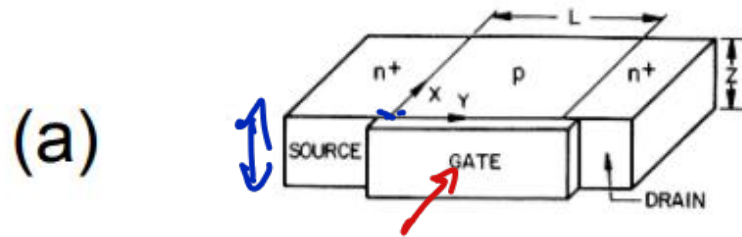
Device (limited to S/D regions)



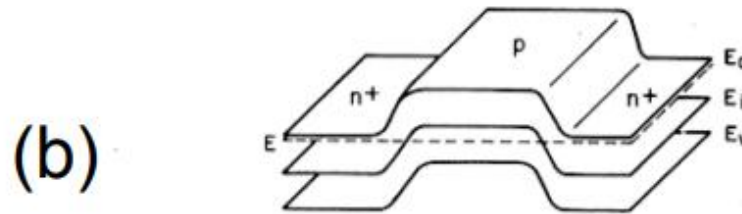
At equilibrium ($V_{GS} = 0, V_{DS} = 0$)



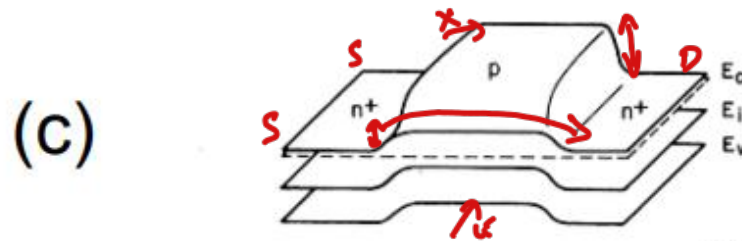
Recap: MOSFET Physics – Band Diagrams



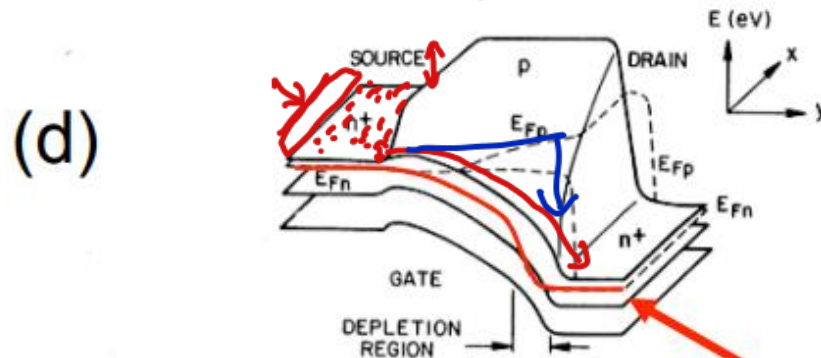
a) device



b) equilibrium (flat band)

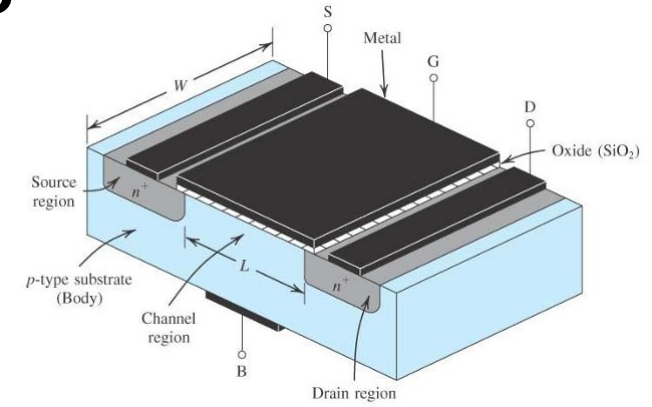


c) equilibrium ($\psi_S > 0$)

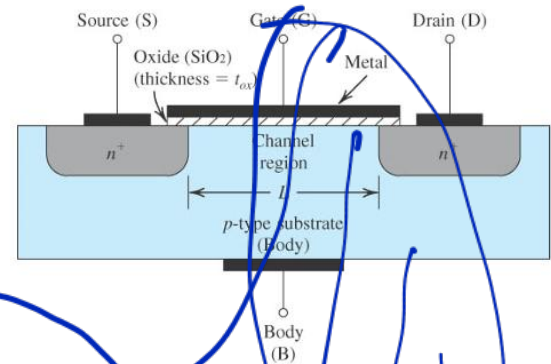


d) non-equilibrium with V_G and $V_D > 0$ applied

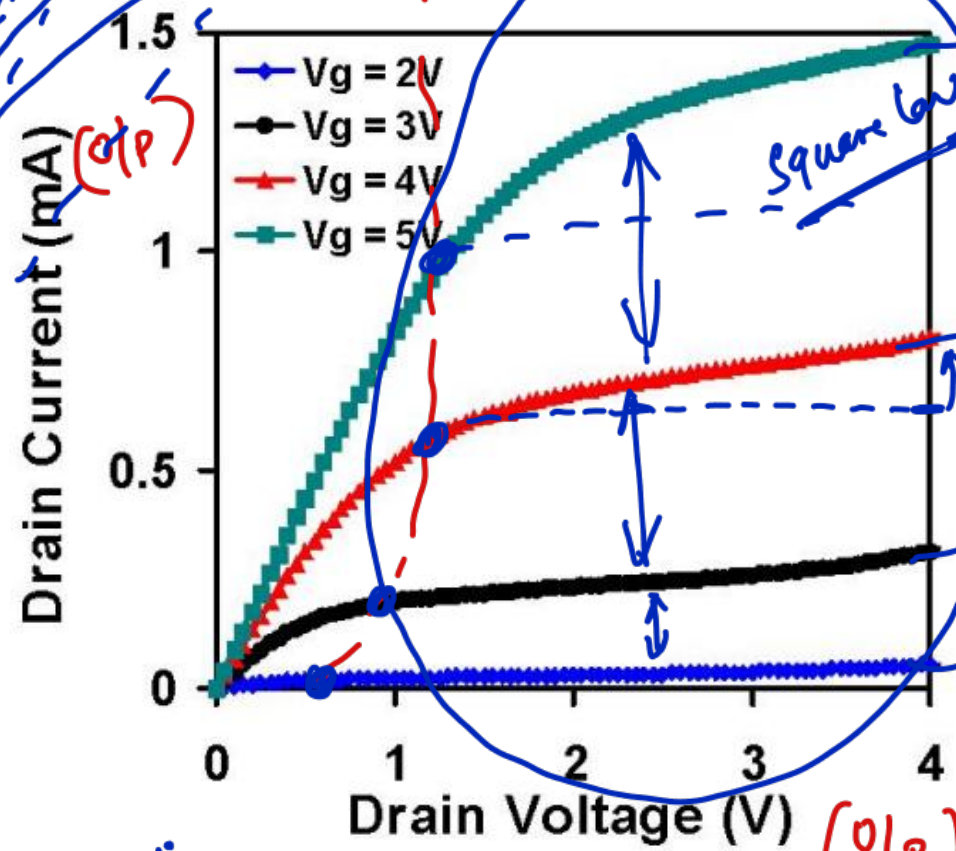
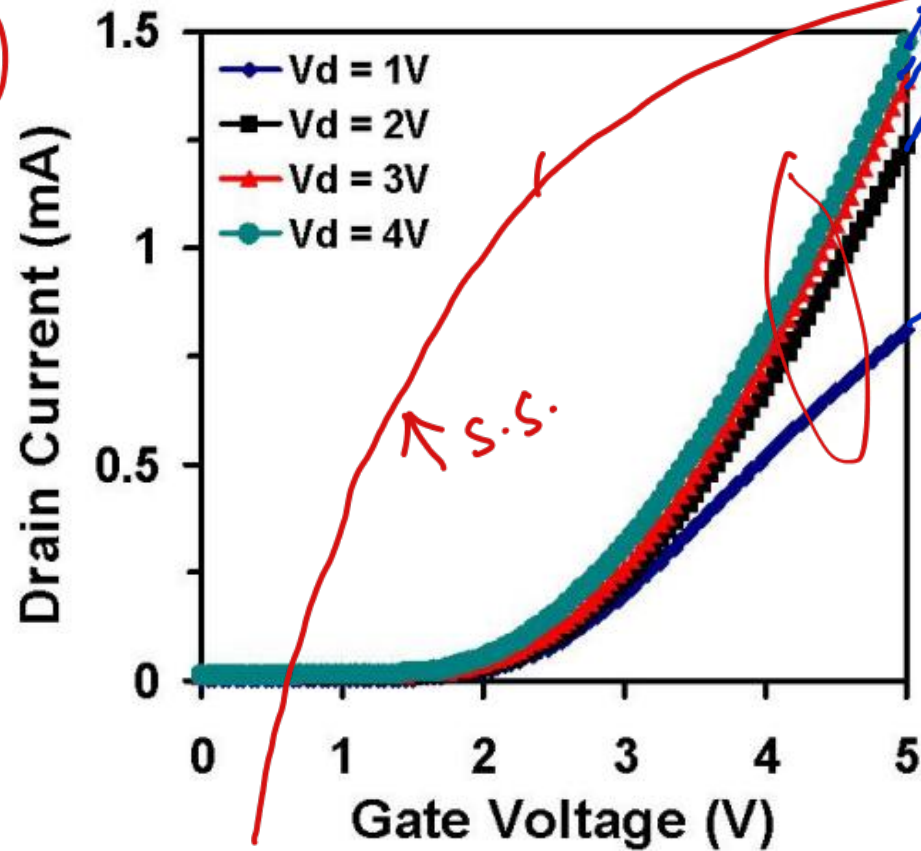
F_N



Recap: MOSFET Operation - Summary



(o/p)



$$I_d = \mu_n C_{ox} \frac{W}{L} \left((V_{GS} - V_{th}) V_{DS} - \frac{V_{DS}^2}{2} \right)$$

$$V_{DS} = V_{GS} - V_{th}$$

MOSFET – How to improve performance?

Ideal Switch vs MOSFET Switch

"OFF"

$$S.S. = V_T \ln(10) \left(1 + \frac{C_{dep}}{C_{ox}}\right)$$

↓ Power ✓

"ON"

$$I_{DS} = \mu_n C_{ox} \frac{W}{L} \left[(V_{GS} - V_{TH}) V_{DS} - \frac{V_{DS}^2}{2} \right]$$

↑ Speed ✓

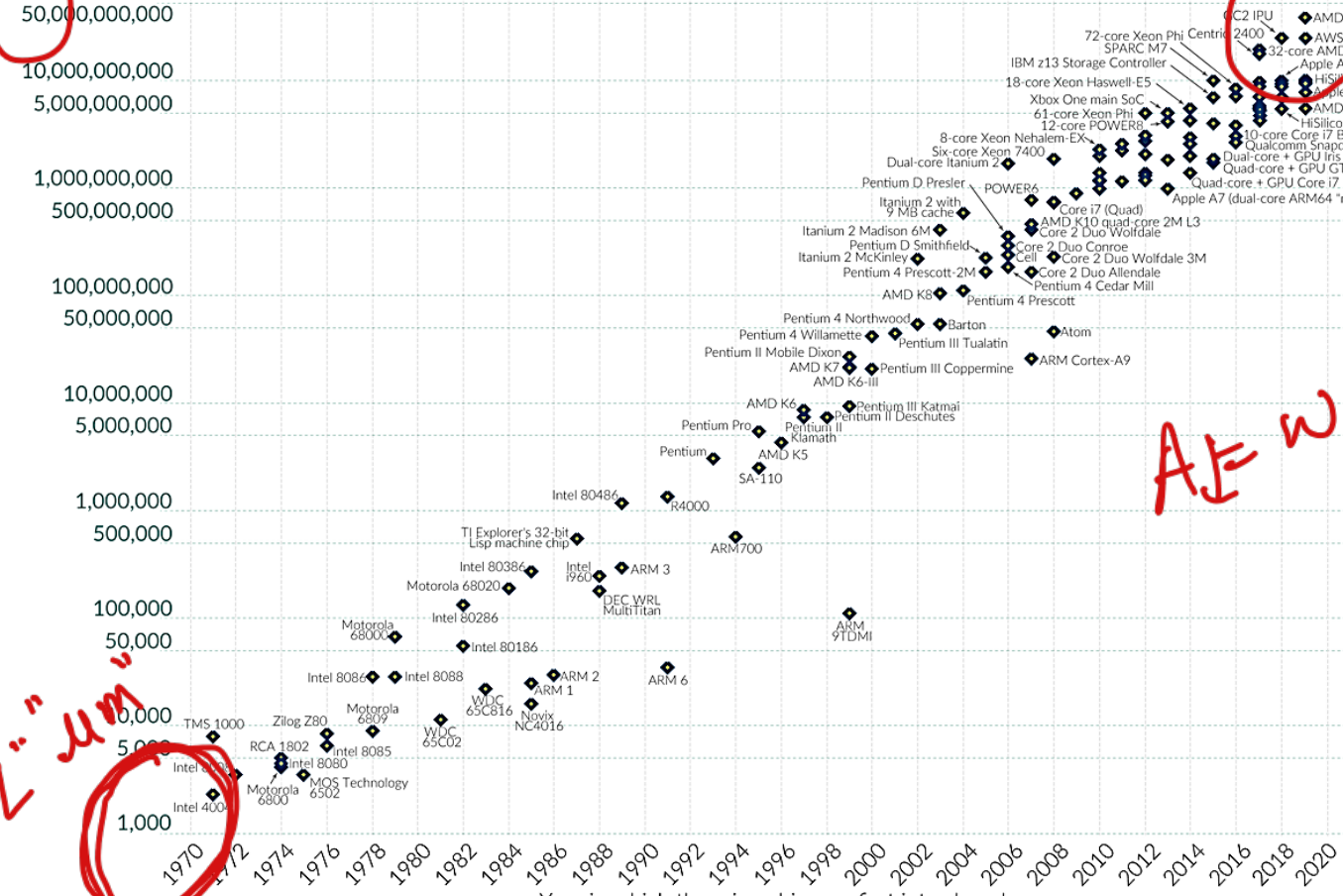
$\mu \uparrow, C_{ox} \uparrow, W \uparrow,$ $L \downarrow,$ $V_{TH} \downarrow$
limitation ✓ limitation

MOSFET Scaling – What is the need?

Moore's Law: The number of transistors on microchips doubles every two years

Moore's law describes the empirical regularity that the number of transistors on integrated circuits doubles approximately every two years. This advancement is important for other aspects of technological progress in computing – such as processing speed or the price of computers.

Transistor count



Data source: Wikipedia (wikipedia.org/wiki/Transistor_count)

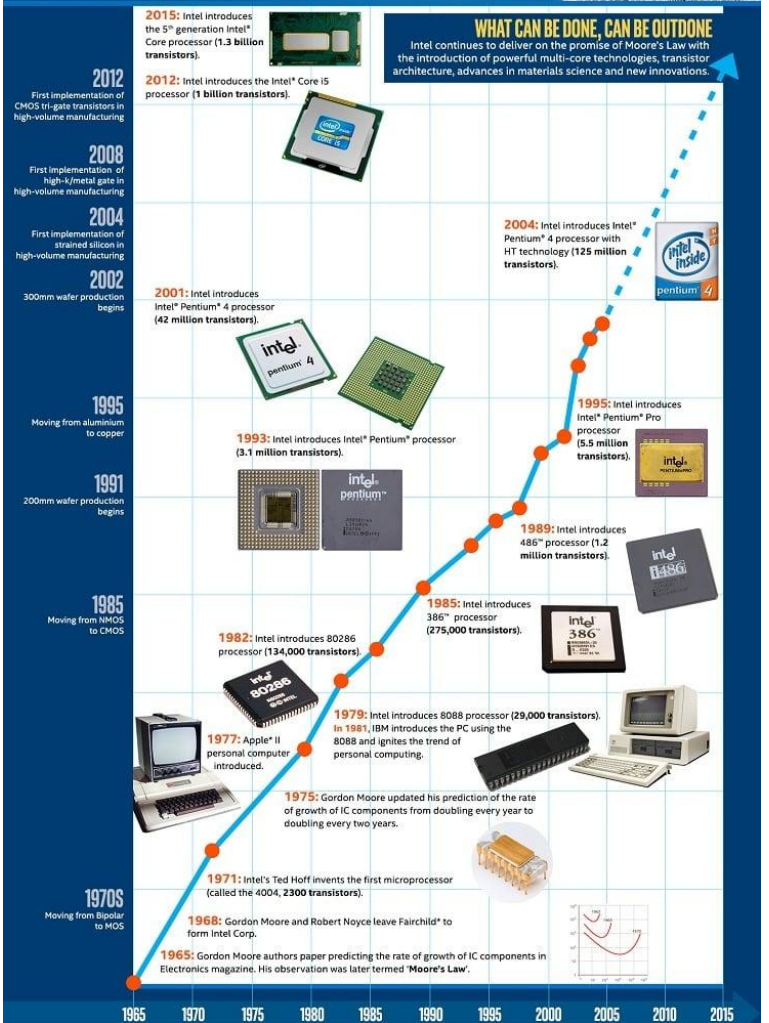
OurWorldinData.org – Research and data to make progress against the world's largest problems.

Licensed under CC-BY by the authors Hannah Ritchie and Max Roser.

Our World
in Data

MOORE'S LAW TIMELINE

Moore's Law – the observation that computing dramatically decreases in cost at a regular pace – is short-hand for rapid technological change. Over the past 50 years, it has ushered in the dawn of the personalization of technology and enabled new experiences through the integration of technology into almost all aspects of our lives.



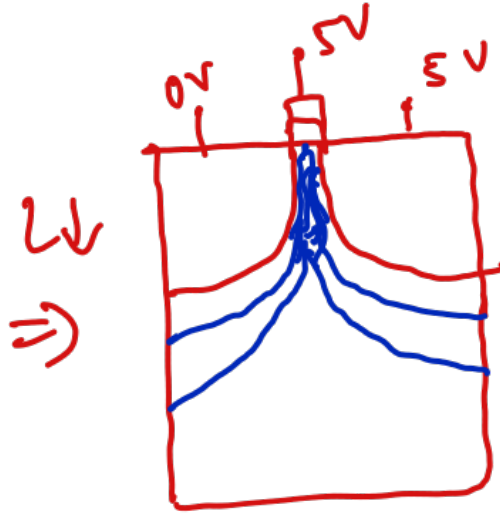
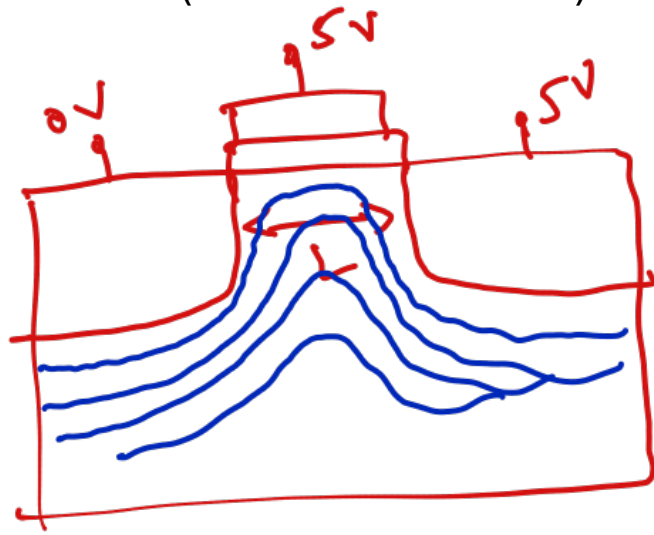
For more information, please visit [intel.com](https://www.intel.com).

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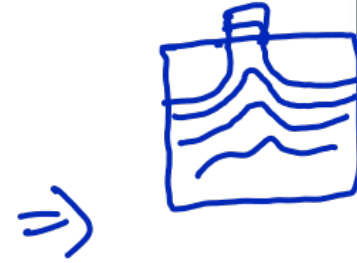


MOSFET Length Scaling – How?

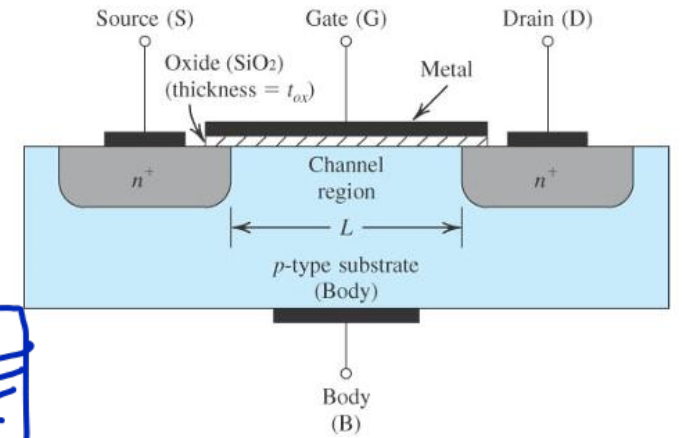
How is E-field (lateral and vertical)?



"L↓"

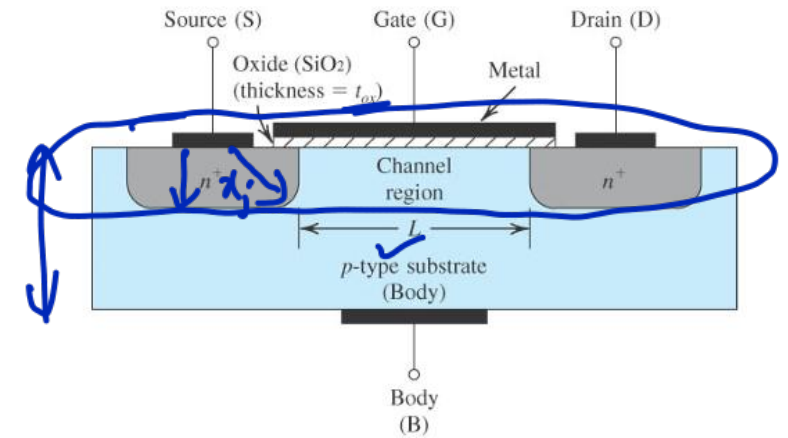


$$E = V/L$$



MOSFET Length Scaling & limitations

Increase in static/standby power consumption due to V_{TH} scaling limitation



Constant E-Scaling (K) " $K > 1$ "

Parameter
 L, W, t_{ox}, x_j ; $L/K, \textcircled{W/K}, t_{ox}/K, x_j/K$

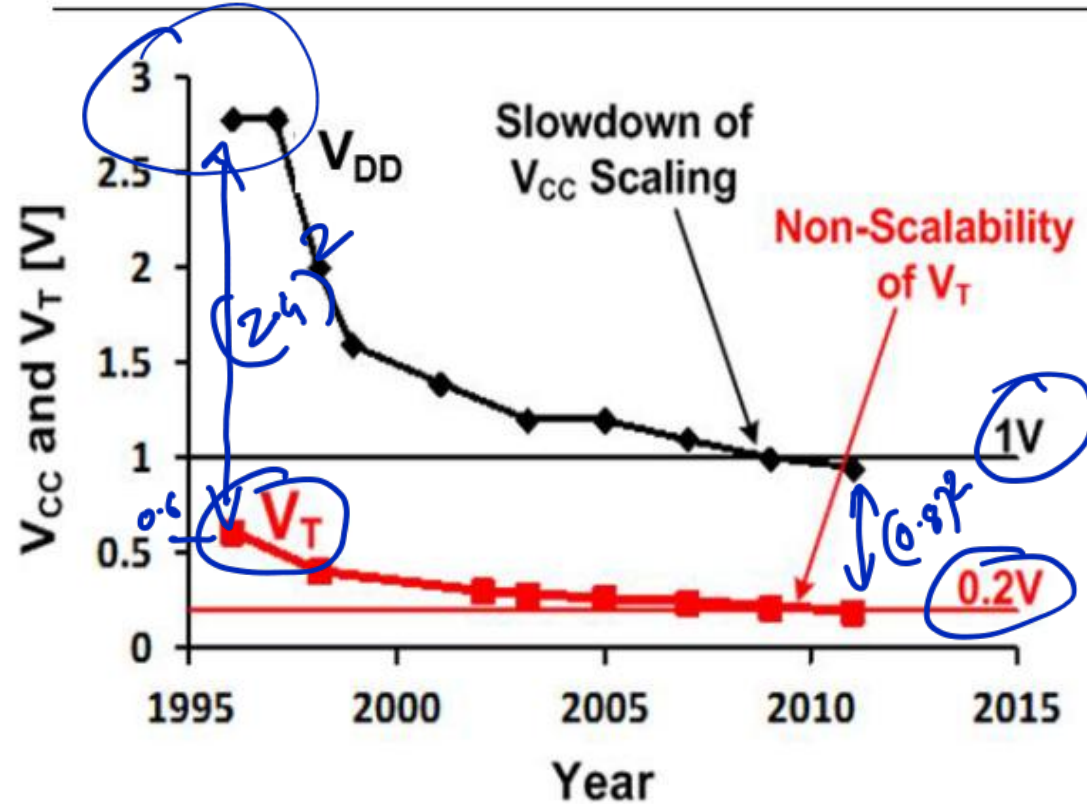
Dopings
 N_D, N_A ; $N_D \cdot K, N_A \cdot K$

Voltages
 $V_{GS}, V_{DS}, (V_{DD})$; $\underline{\underline{V_{DD}/K}}$ ($K > 1$)

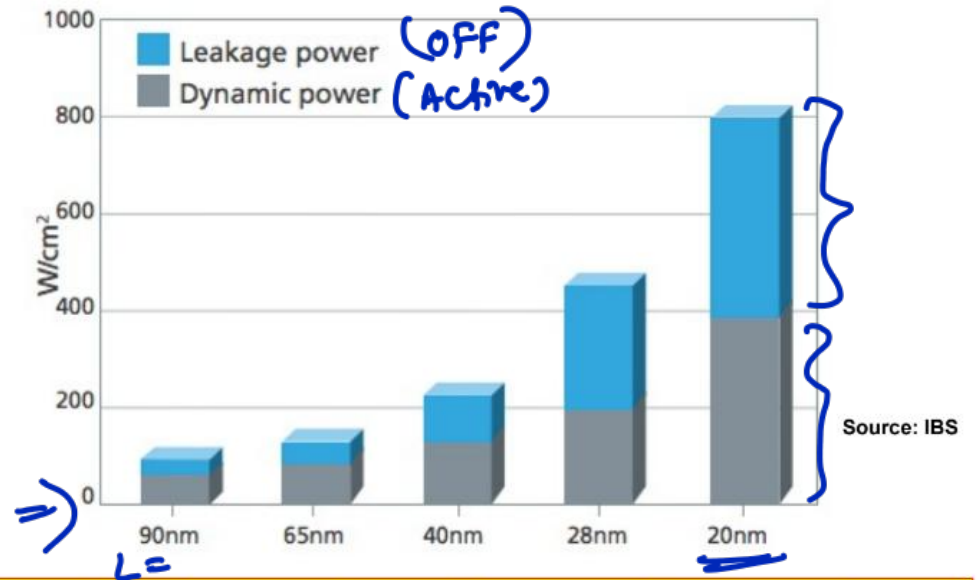
$V_{TH} \uparrow$; N_A
 $2\phi_0 + \sqrt{\quad}$

MOSFET Length Scaling & limitations $\propto (V_{GS} - V_{TH})^2$

Voltage scaling

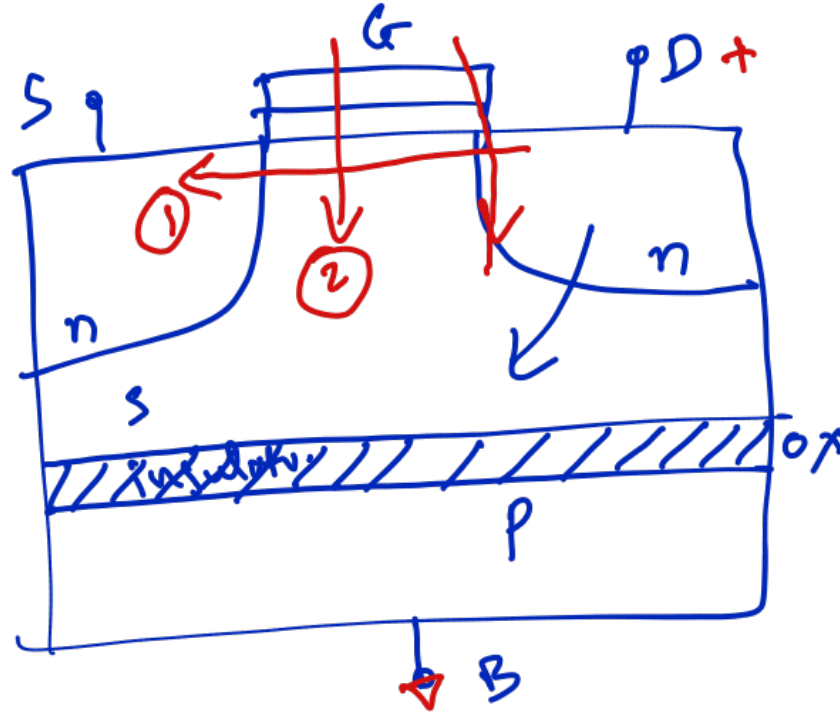


Scaling – The Power Problem



Leakage (idle) power becoming equal to dynamic (active) power with technology scaling

MOSFET – Other Leakage paths & mitigation strategies



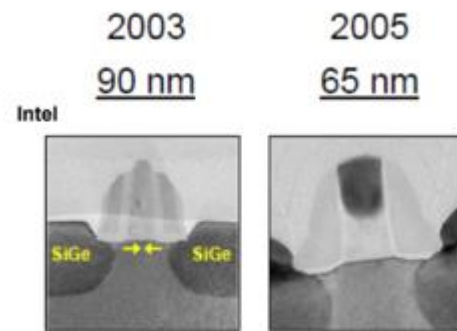
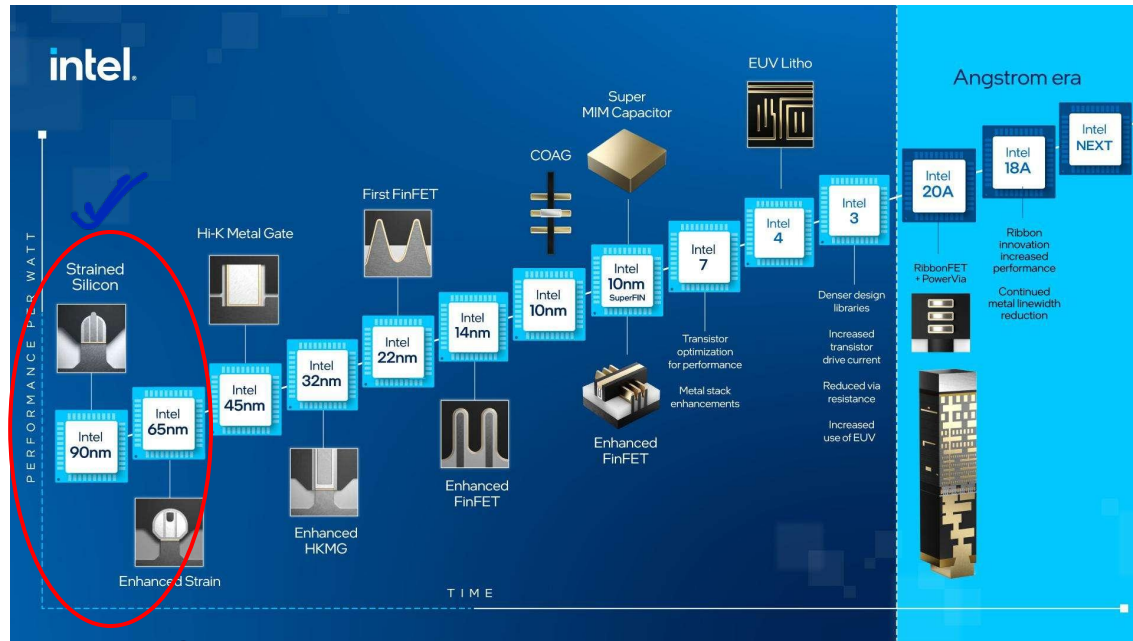
SOI MOSFET

- $\epsilon_{ox} \downarrow$ $C_{ox} \uparrow$
 $C_{dep} \downarrow, C_{ox} \uparrow$
- 1) S.S. leakage
 - 2) Gate leakage (tunneling) $\epsilon_{ox} \uparrow$
 - 3) Body leakage (P-B diode) SOI FET
 - 4) GIDL (Gate induced drain leakage)

MOSFET – Other Leakage paths & mitigation strategies

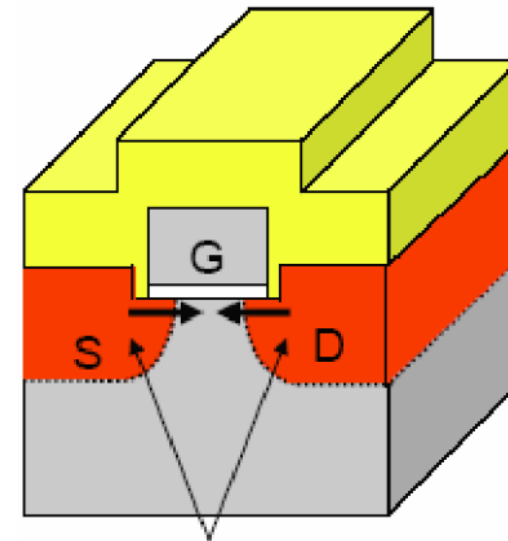
" $\mu \uparrow$ $C_{ox} \uparrow$ " $L \downarrow$


MOSFET Length Scaling – Increasing Mobility



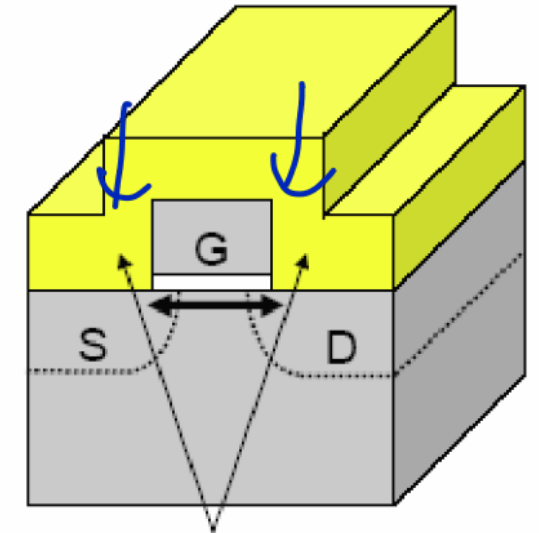
Strain - mobility

Intel's 90nm Technology



Selective SiGe S-D

PMOS



Tensile Si₃N₄ Cap

NMOS

MOSFET – Gate Leakage reduction strategy

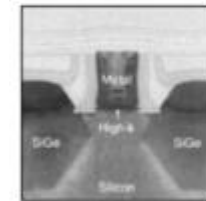
- Use High-k dielectric

$t_{ox} \downarrow$ $C_{ox} \uparrow = \frac{k\epsilon}{t_{ox}}$

How to reduce oxide thickness without too much leakage (e.g. tunnelling leakage)

4,30

2007
45 nm



2009
32 nm



HKMG - leakage

Effective oxide thickness (EOT)

- With down-scaling, need to maintain minimum gate capacitance C
- EOT gives thickness of SiO_2 dielectric film required to reach a certain capacitance density (C/A)

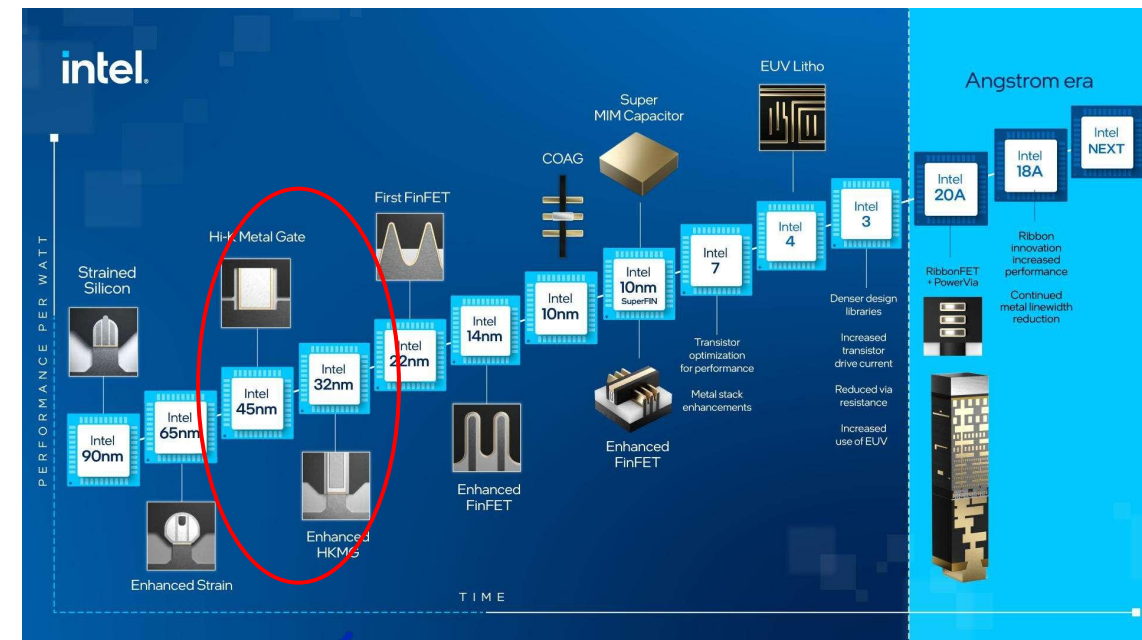
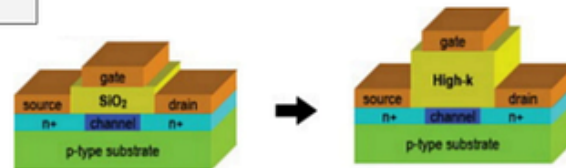
$$C = \kappa \epsilon_0 \frac{A}{t} \quad \downarrow$$

Maintaining C

- 1) Decrease t_{ox}
- 2) Use higher k

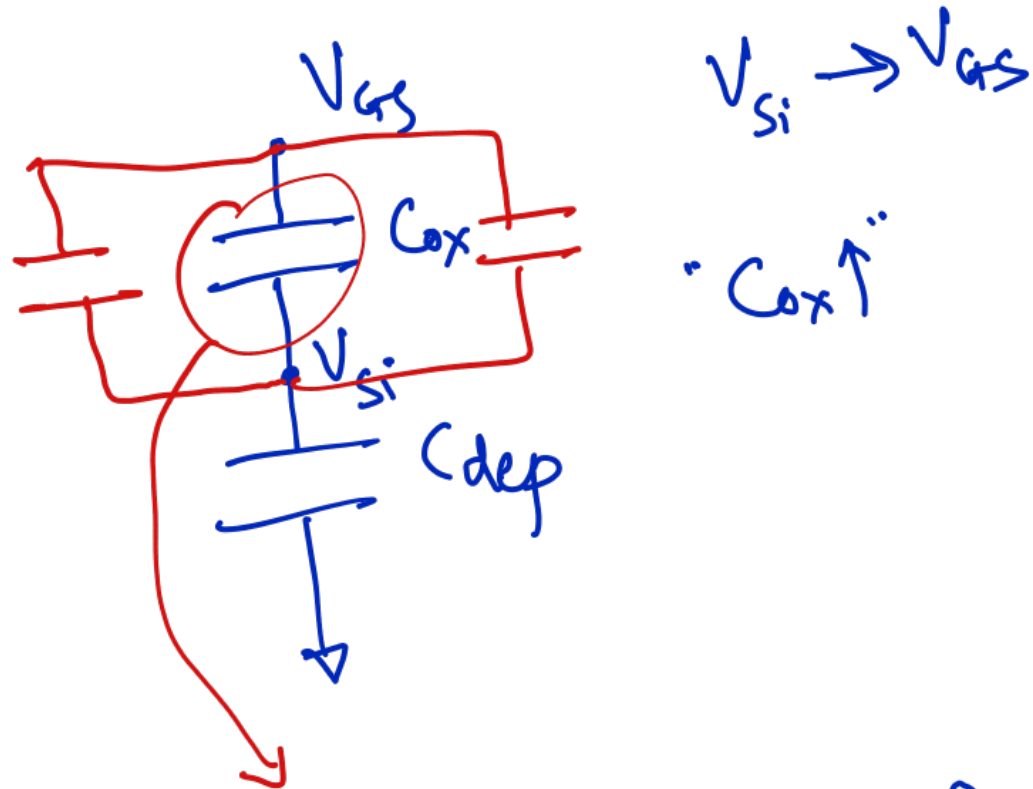
$$\frac{C}{A} = \frac{\kappa_{\text{SiO}_2} \epsilon_0}{t_{\text{SiO}_2}} = \frac{\kappa_{hk} \epsilon_0}{t_{hk}}$$

$$EOT = t_{ox} = t_{hk} \frac{\kappa_{\text{SiO}_2}}{\kappa_{hk}}, \quad \kappa_{\text{SiO}_2} = 3.9$$

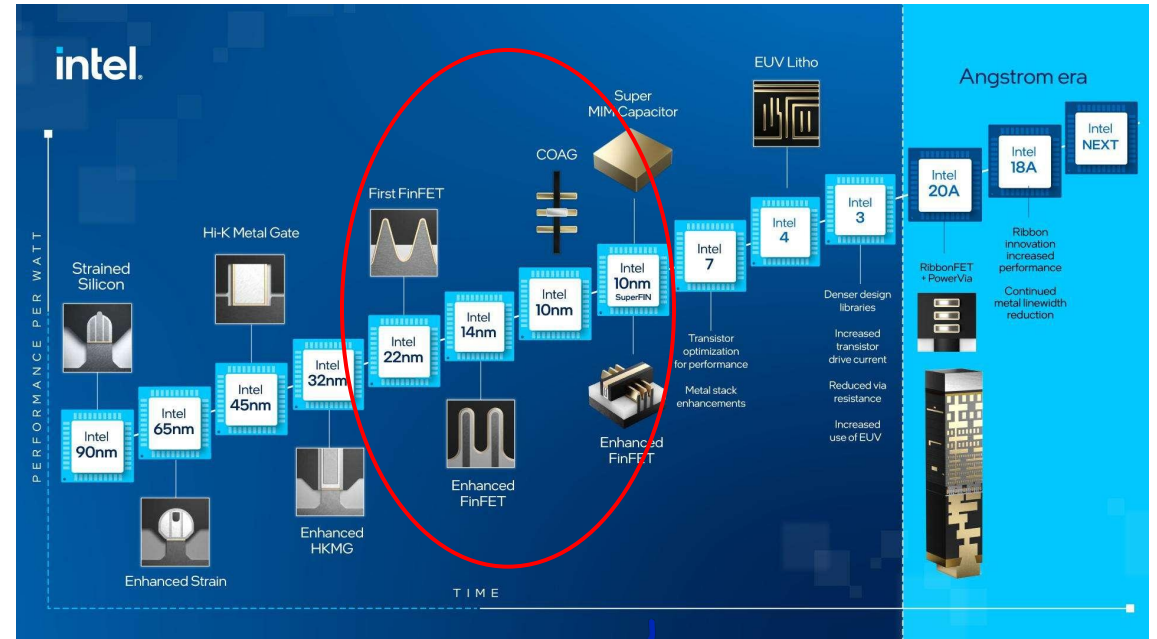
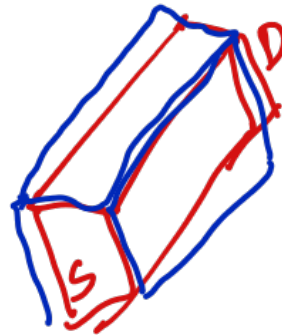
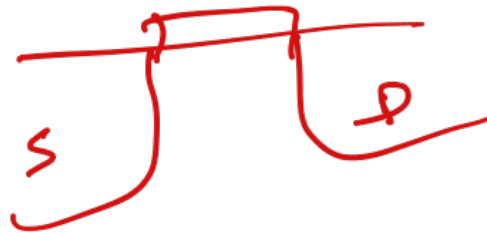


$\mu \uparrow$ High k $C_{ox} \uparrow$

MOSFET Length Scaling – Increasing Coupling (C_{ox})



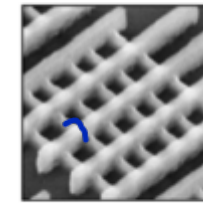
$V_{Si} \rightarrow V_{GS}$
" $C_{ox} \uparrow$ "



intel

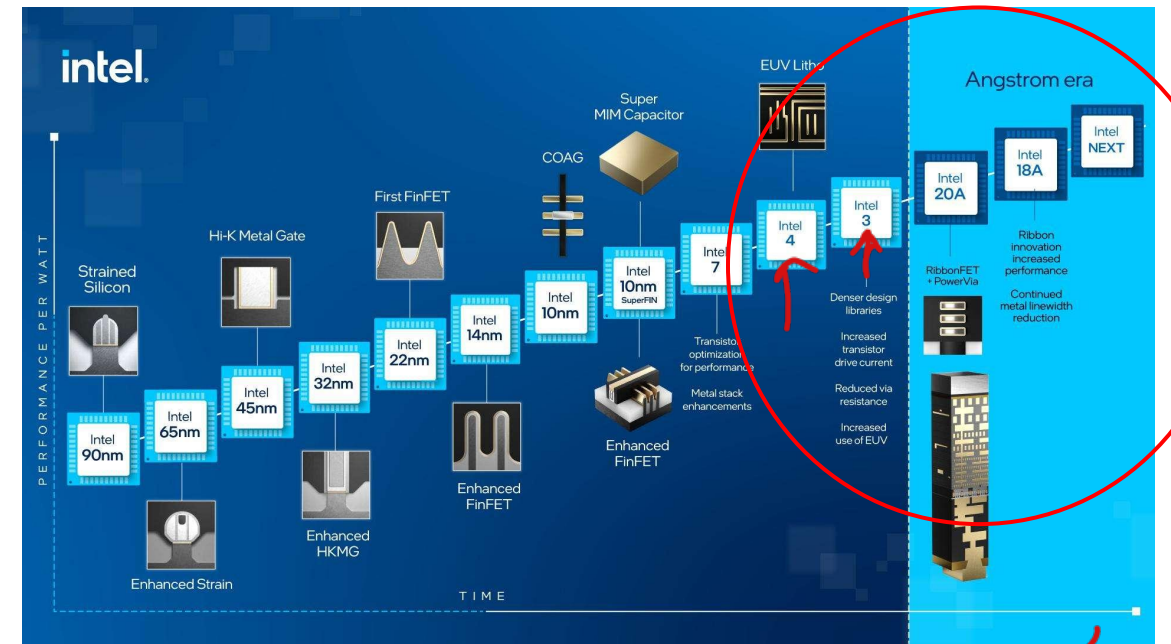
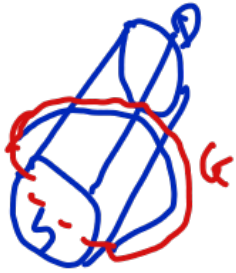
2011

22 nm

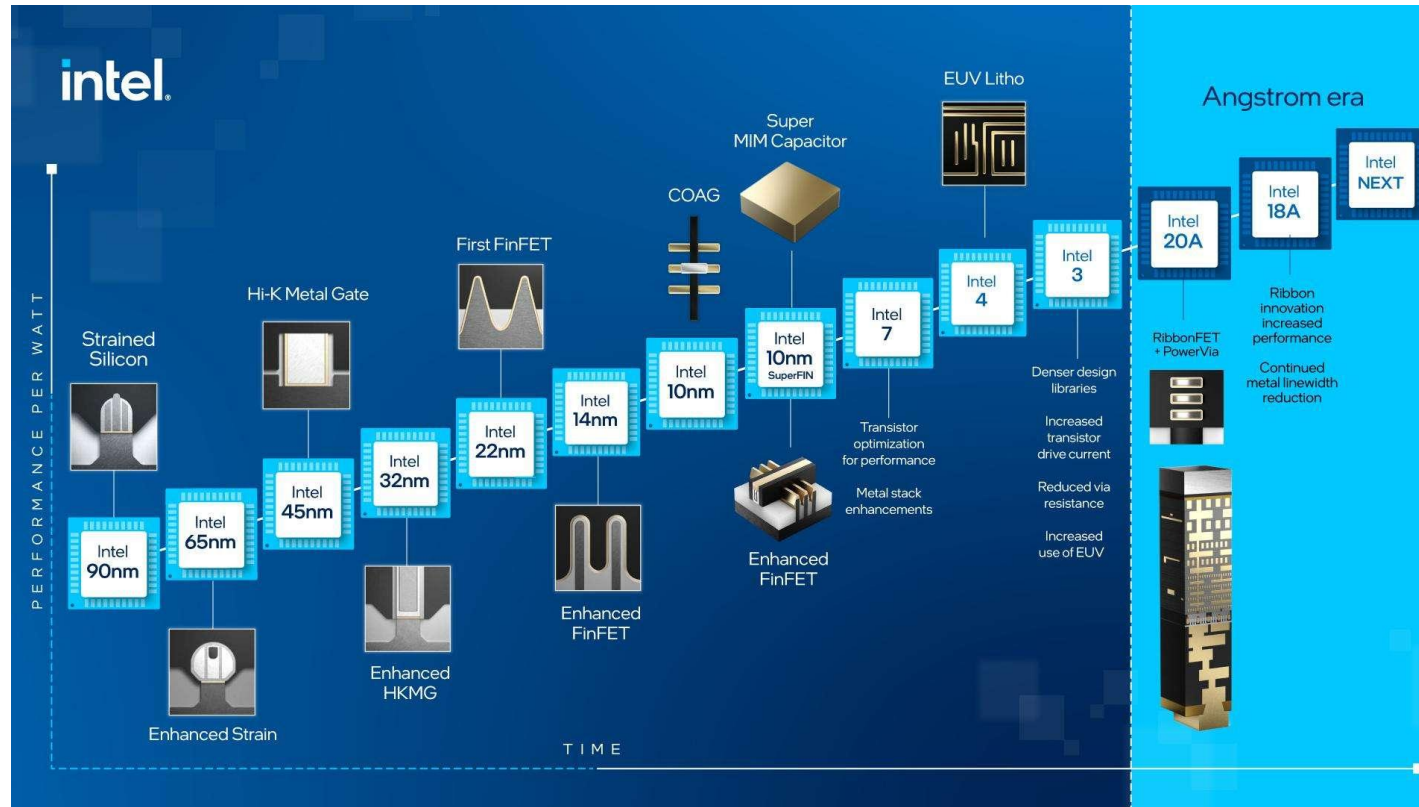


FinFET - SCE

MOSFET Length Scaling – Increasing Coupling (C_{ox})

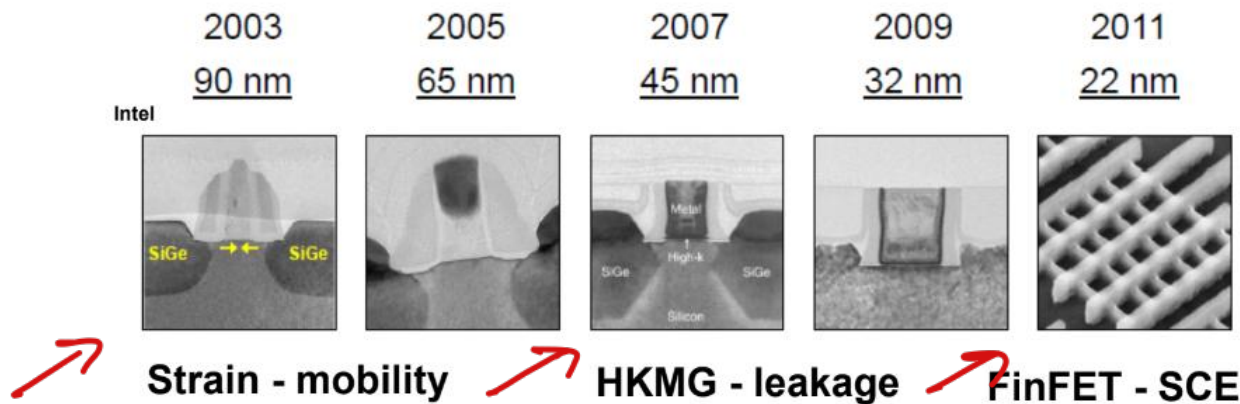


MOSFET Scaling – over the time



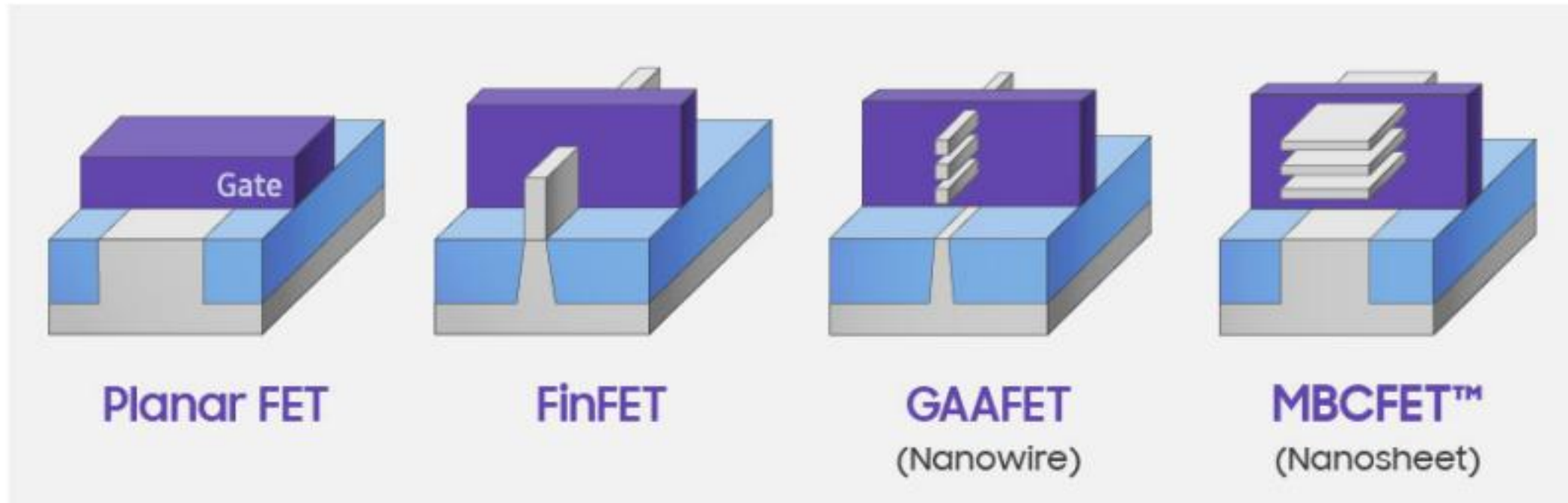
Aspiring to
1 Trillion
transistors in 2030

- ✓ RibbonFET
- ✓ PowerVia
- ✓ High NA
- ✓ 2.5D/3D packaging



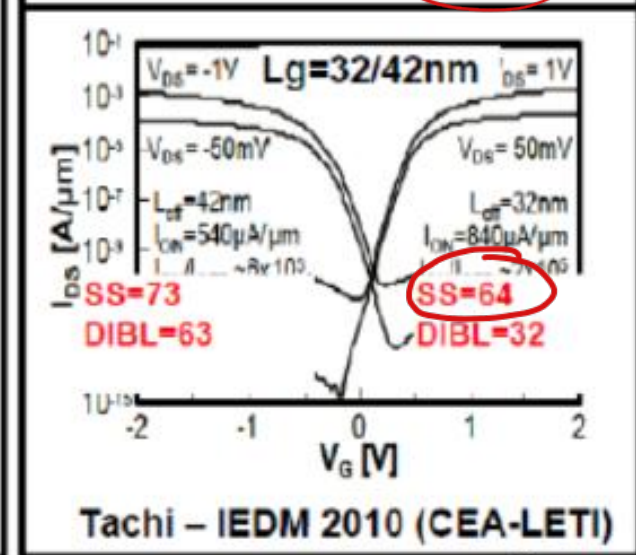
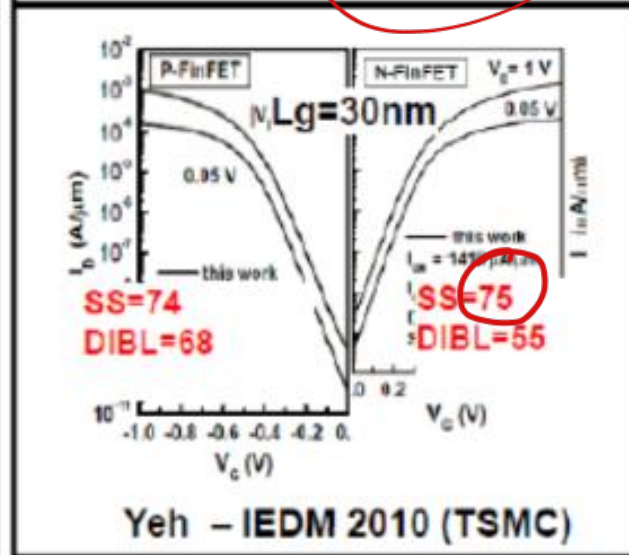
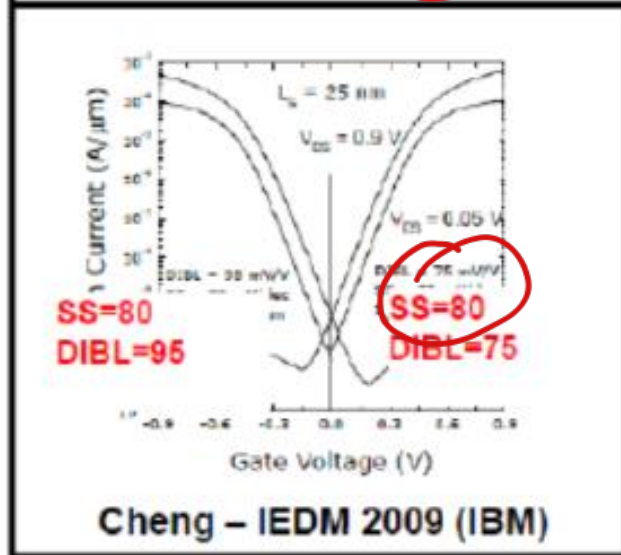
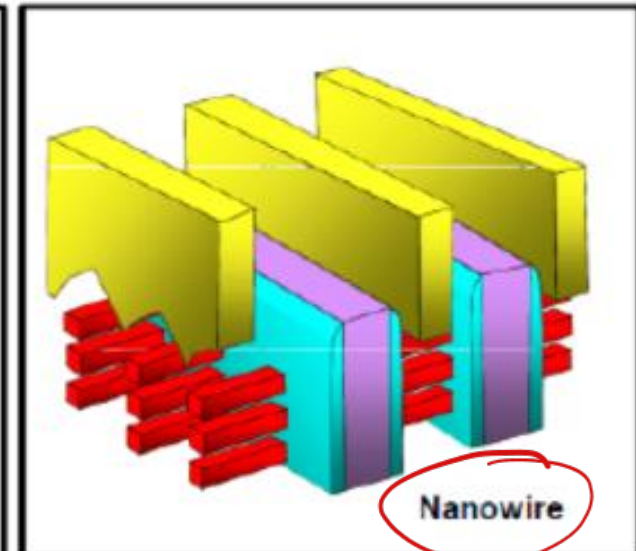
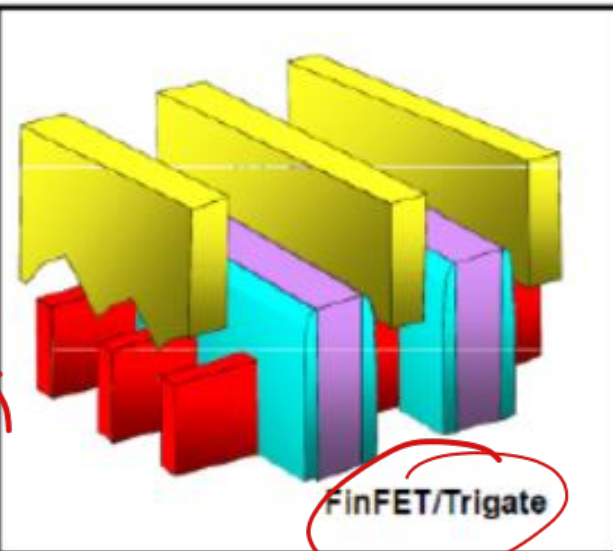
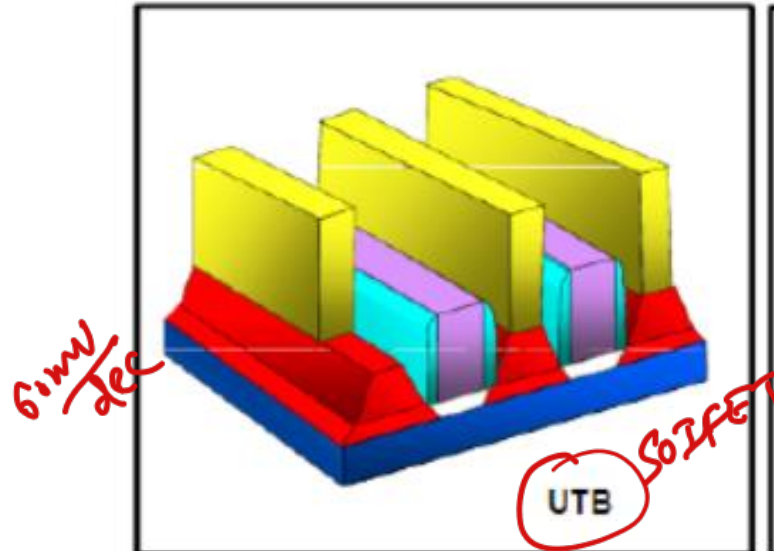
GRAFET

MOSFET Length Scaling – Increasing Coupling



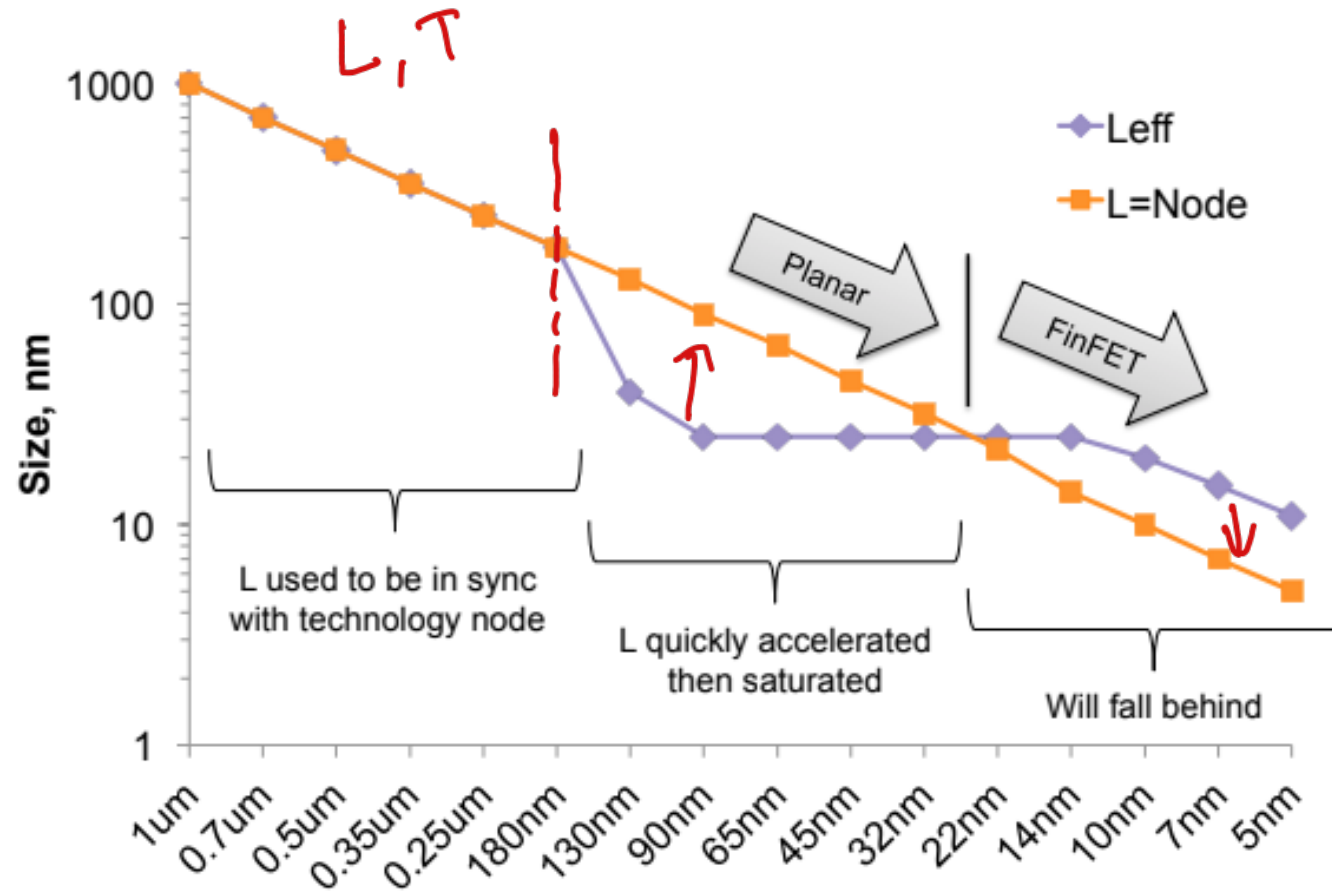
<https://www.youtube.com/watch?v=QAAuO-bfyb0&t=69s>

MOSFET Scaling – More challenges and solutions



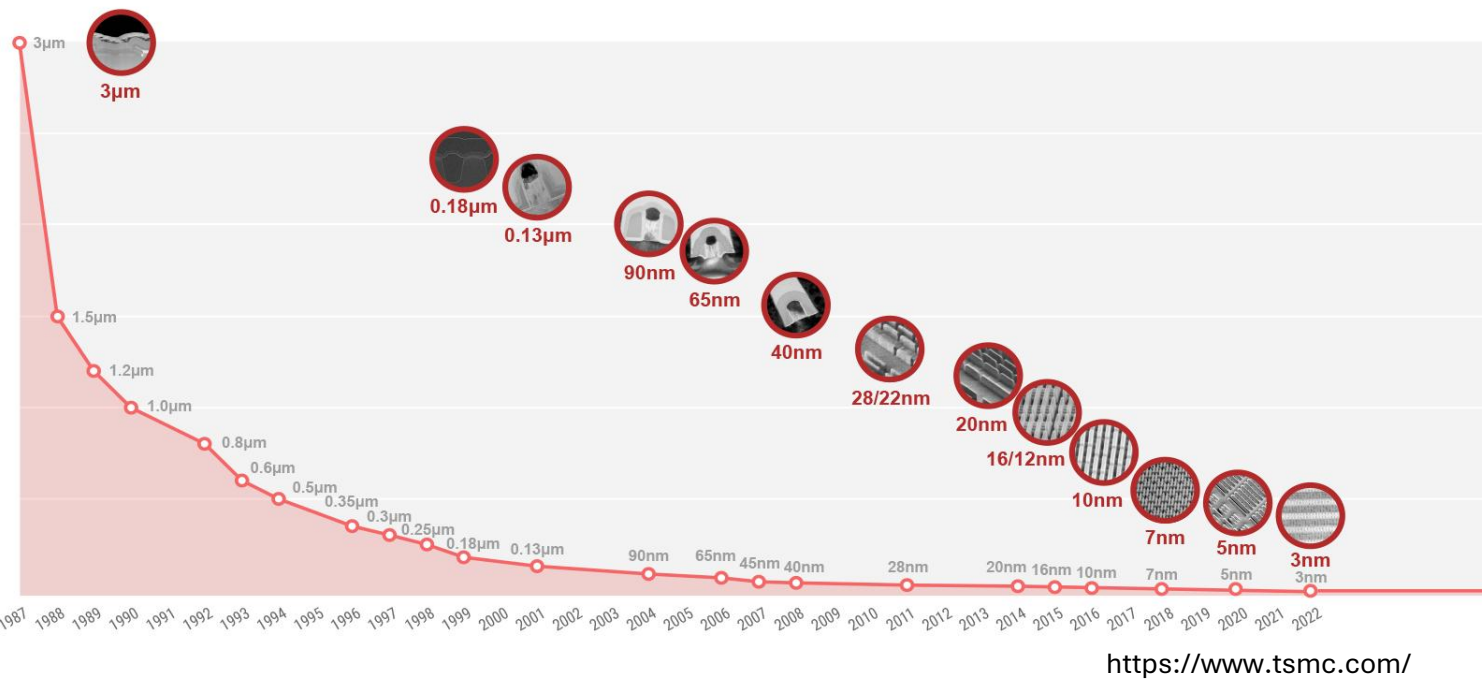
MOSFET Scaling – over the time

Evolution of Transistor Scaling



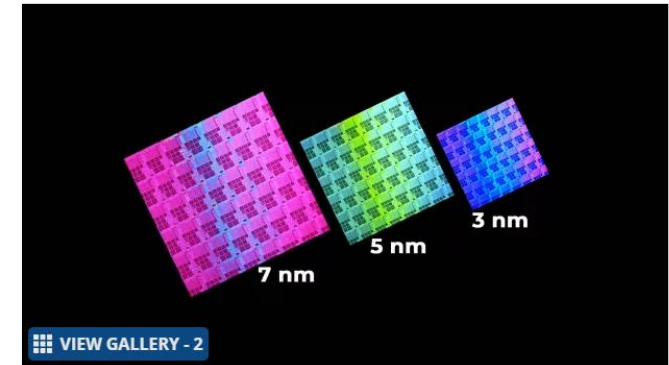
The node numbers (such as “14nm node”) are not related to anything anymore, just an indication of time

MOSFET Scaling – over the time

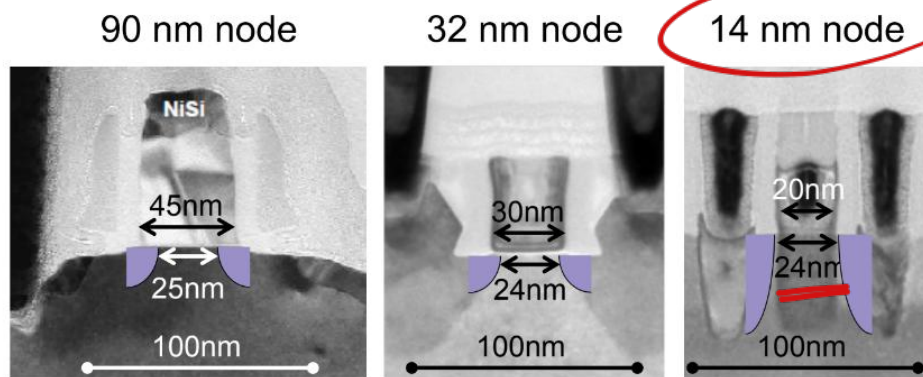


TSMC's 5nm, 3nm production lines remain 'fully loaded' in 1H 2025, strong orders for AI chips

TSMC's 5nm and 3nm production lines are 'fully loaded' until 1H 2025 due to 'strong orders' for AI chips, Qualcomm + MediaTek smartphone chips.



<https://www.tweaktown.com/>

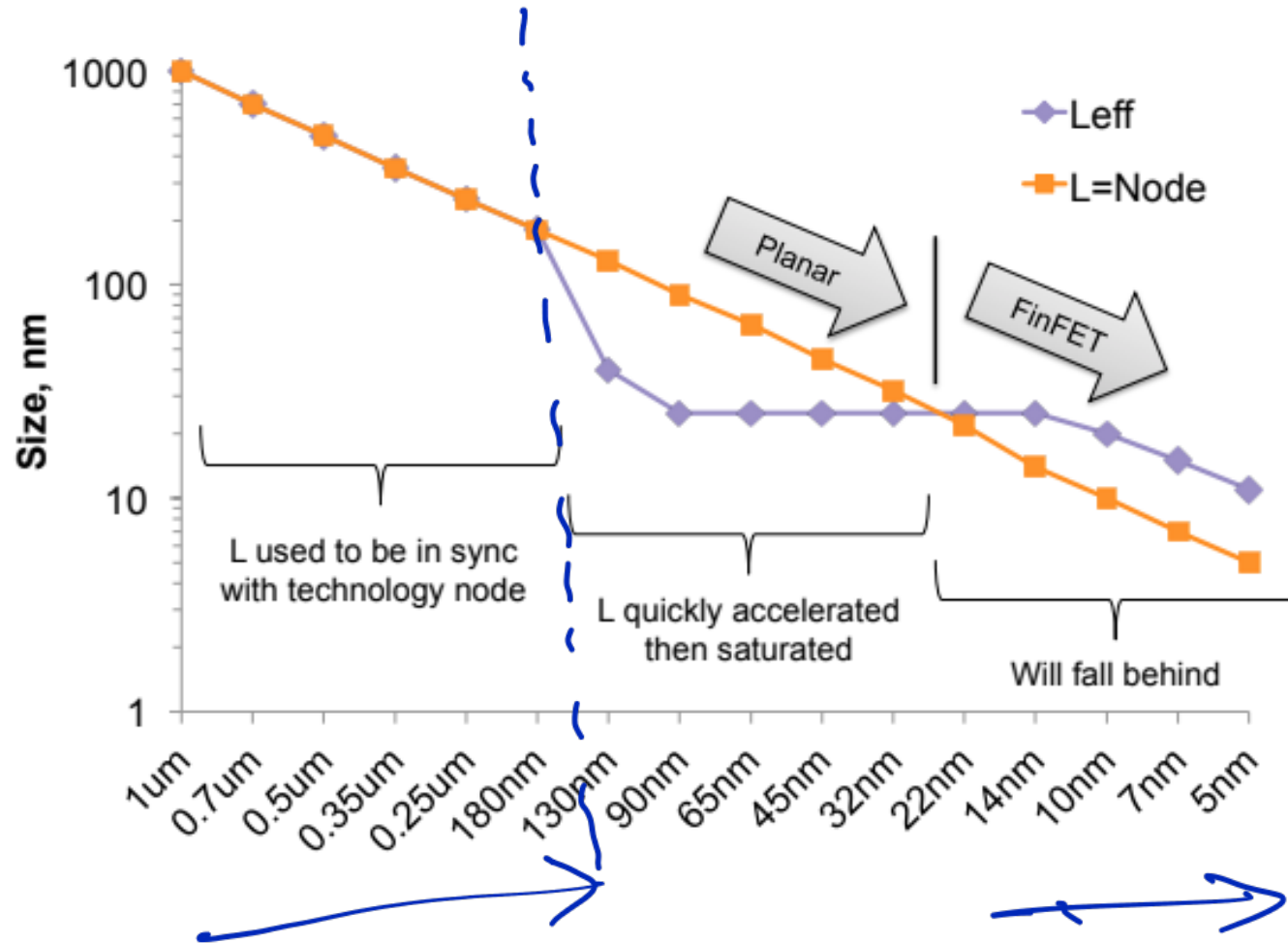


- Lgate shrinks very slow
- S/D overlap shrinks fast and becomes underlap
- Leff stays almost fixed

The technology node numbers (such as “14nm node”) are not related to anything anymore, just an indication of time

MOSFET Scaling – over the time

Evolution of Transistor Scaling

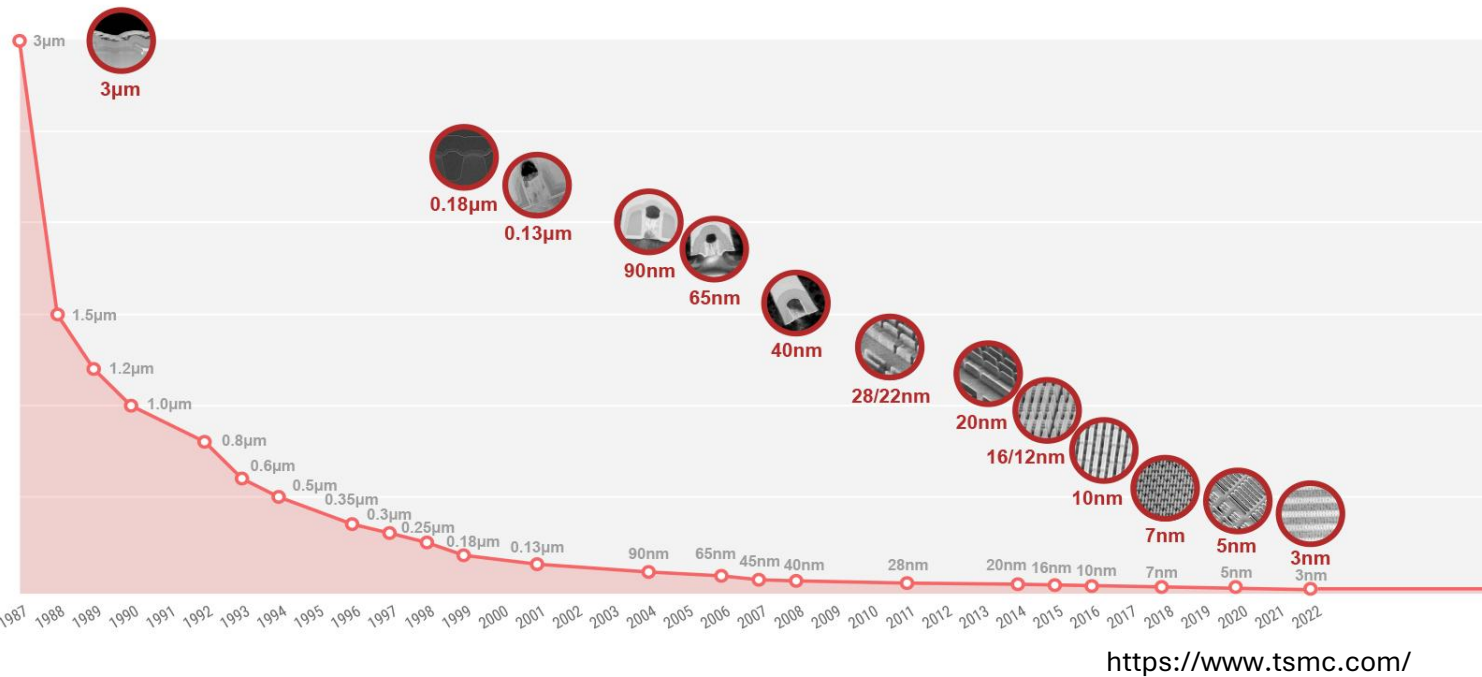


$$\uparrow I_{on} \propto \frac{1}{L} \downarrow$$

The node numbers (such as “14nm node”) are not related to anything anymore, just an indication of time

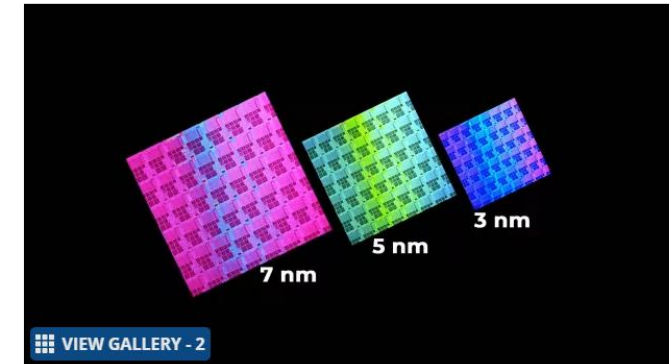
E-field

MOSFET Scaling – over the time

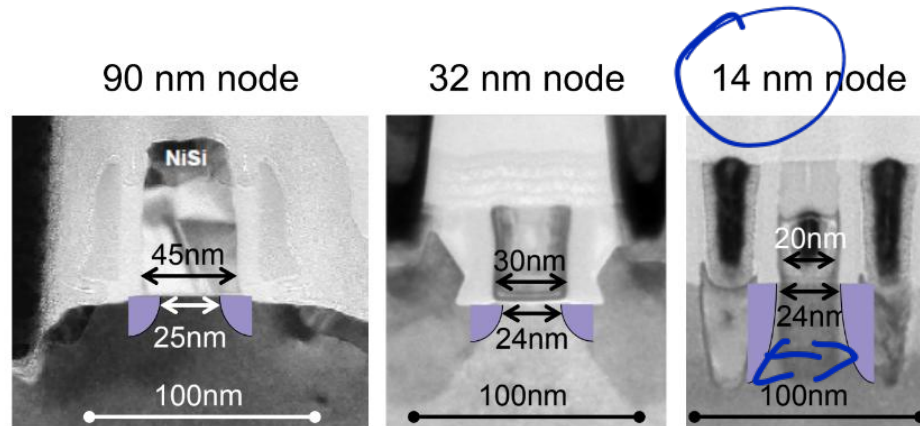


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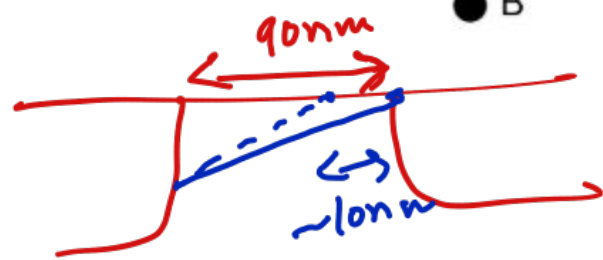
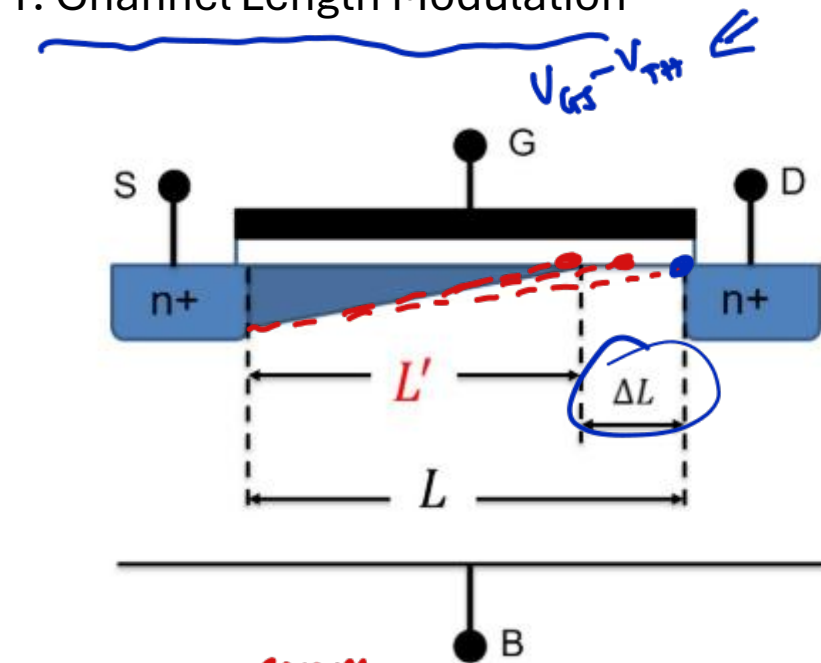


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The technology node numbers (such as “14nm node”) are not related to anything anymore, just an indication of time

MOSFET Scaling – Short channel Effects

1. Channel Length Modulation



$$V_{GS} - V_{TH} = V_{DSat}$$

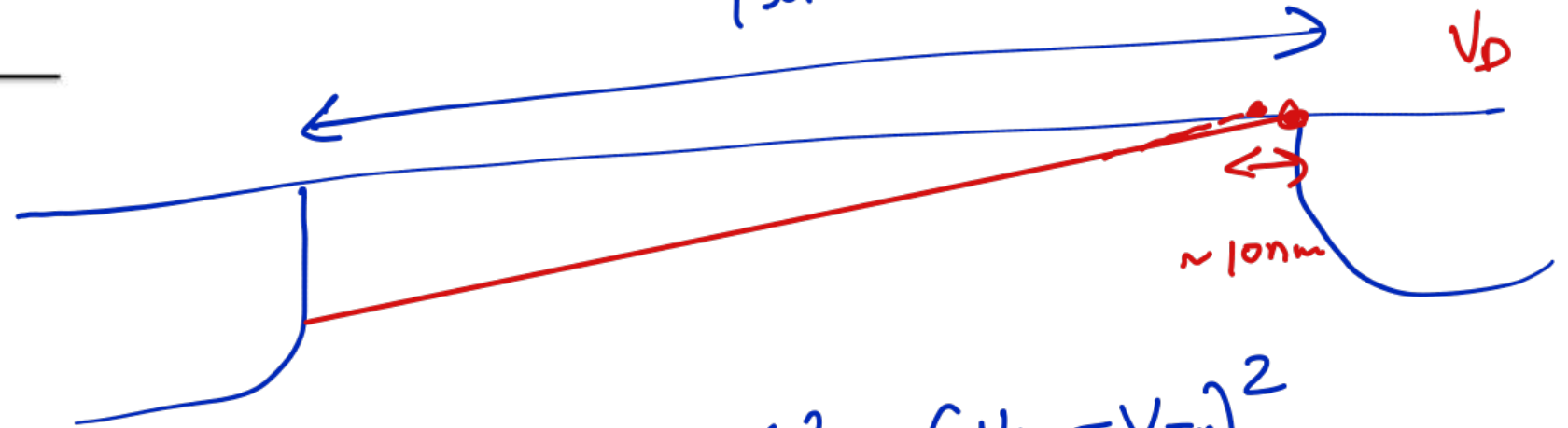
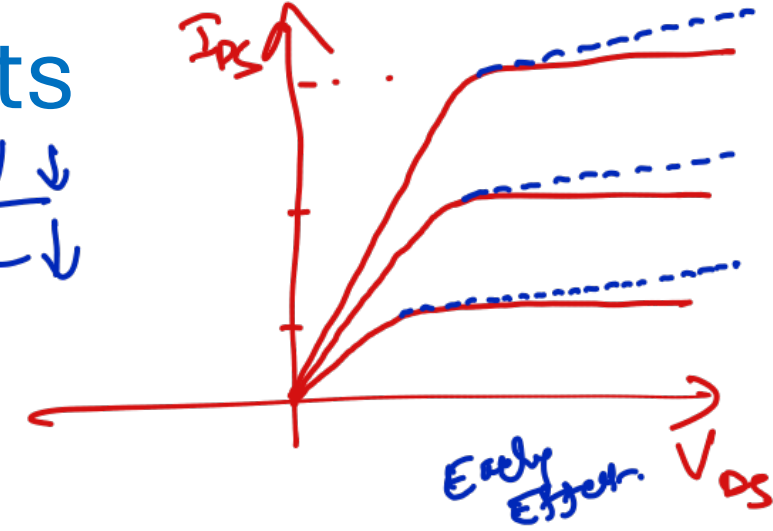
$$E = \frac{V_{DS}}{L}$$

$$I_{ON} \uparrow$$

$$I_{OFF} \downarrow$$

$1\mu\text{m}$

$\approx L_{eff} \sim 1\mu\text{m}$



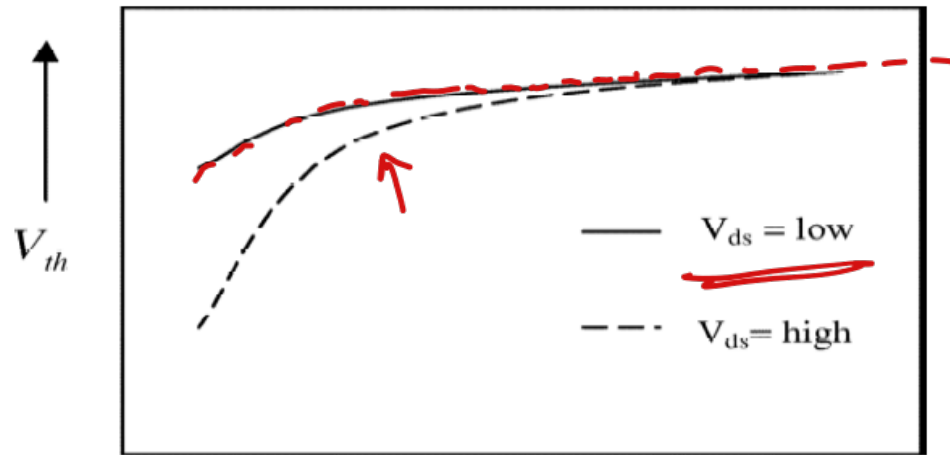
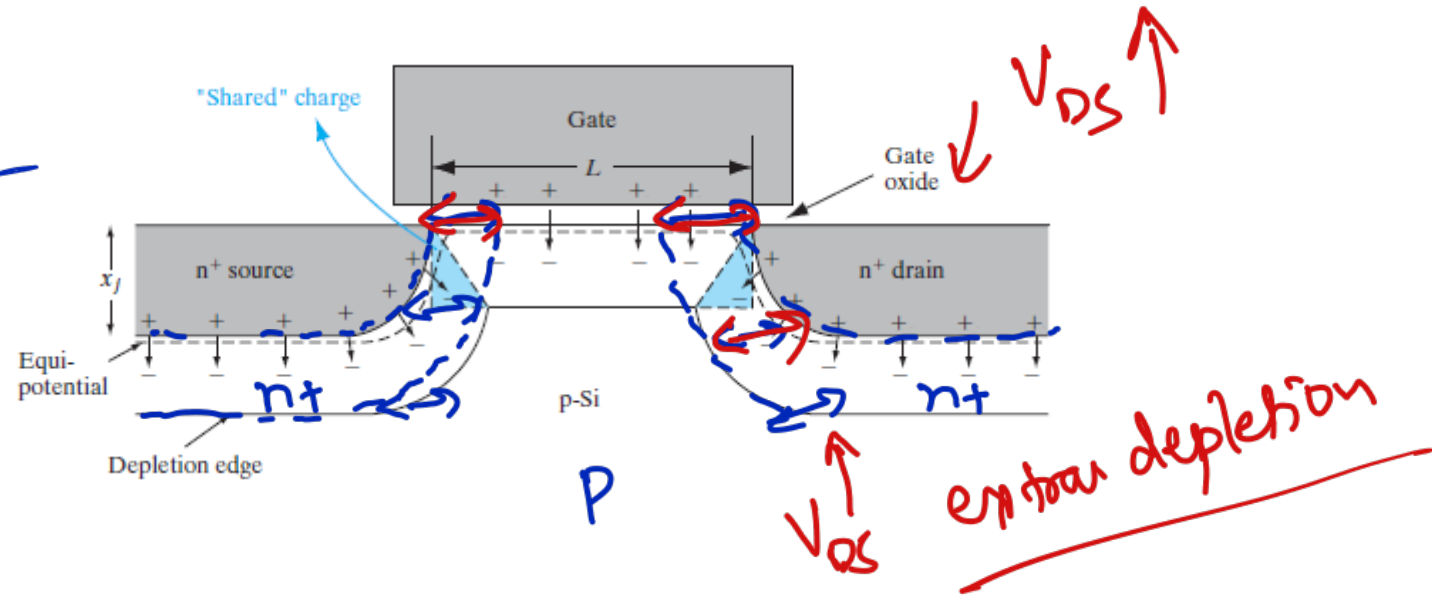
$$I_{ON} = \mu_n C_{ox} \frac{W}{L_{eff}} (V_{GS} - V_{TH})^2$$

$\underbrace{L_{eff}}_{L - \Delta L}$

MOSFET Scaling – Short channel Effects

2. V_{TH} roll off

charge sharing

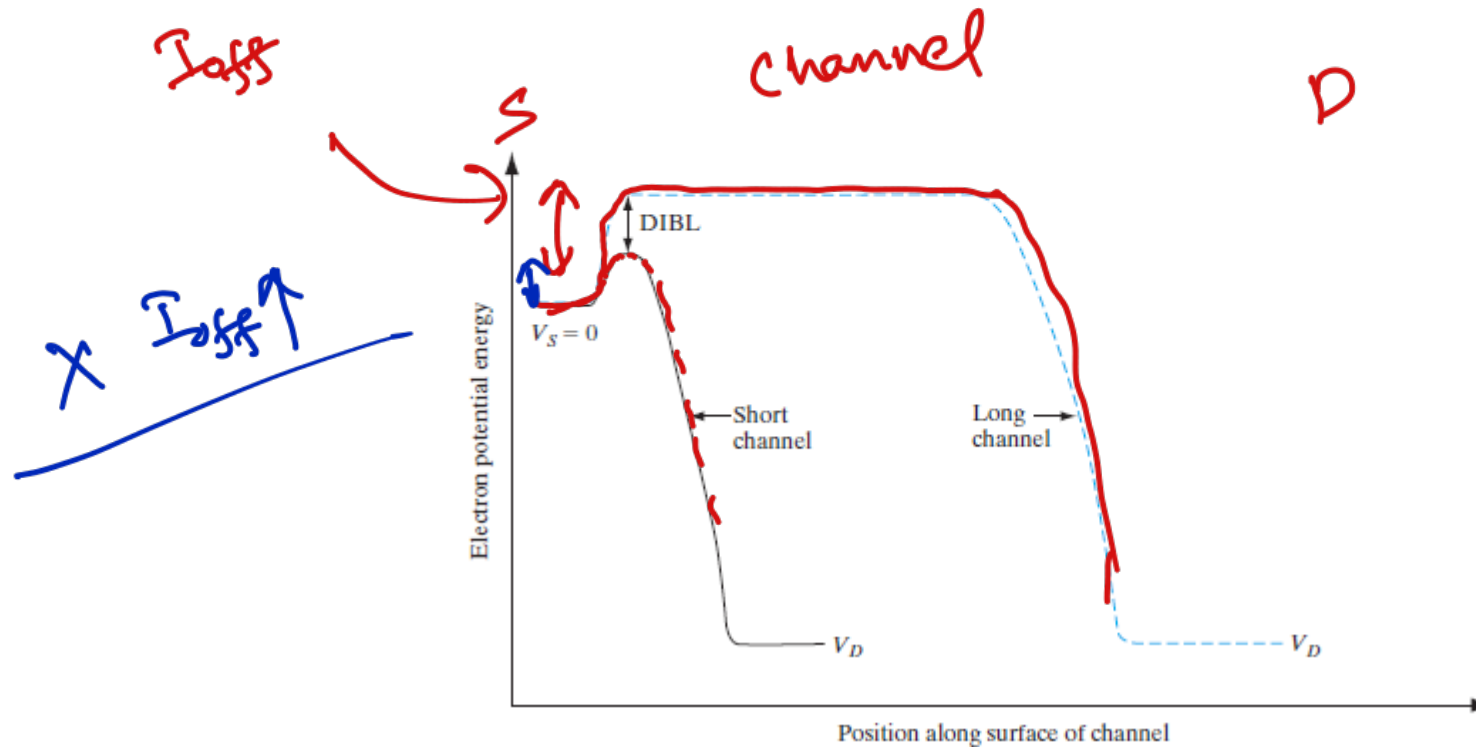
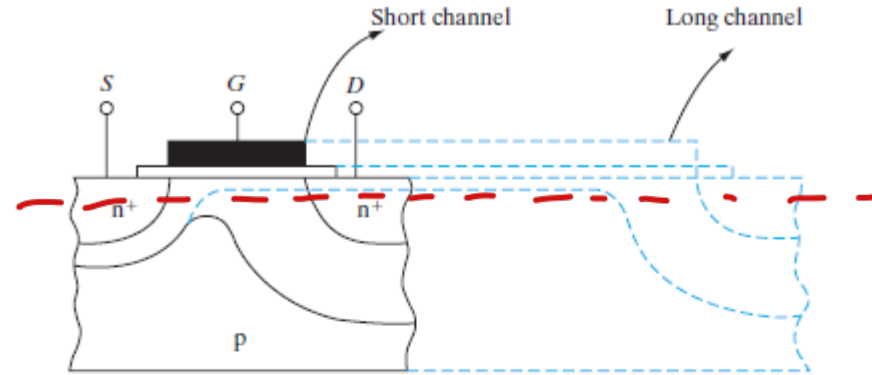


$V_{TH} \downarrow$, $I_{off} \uparrow$ X

MOSFET Scaling – Short channel Effects

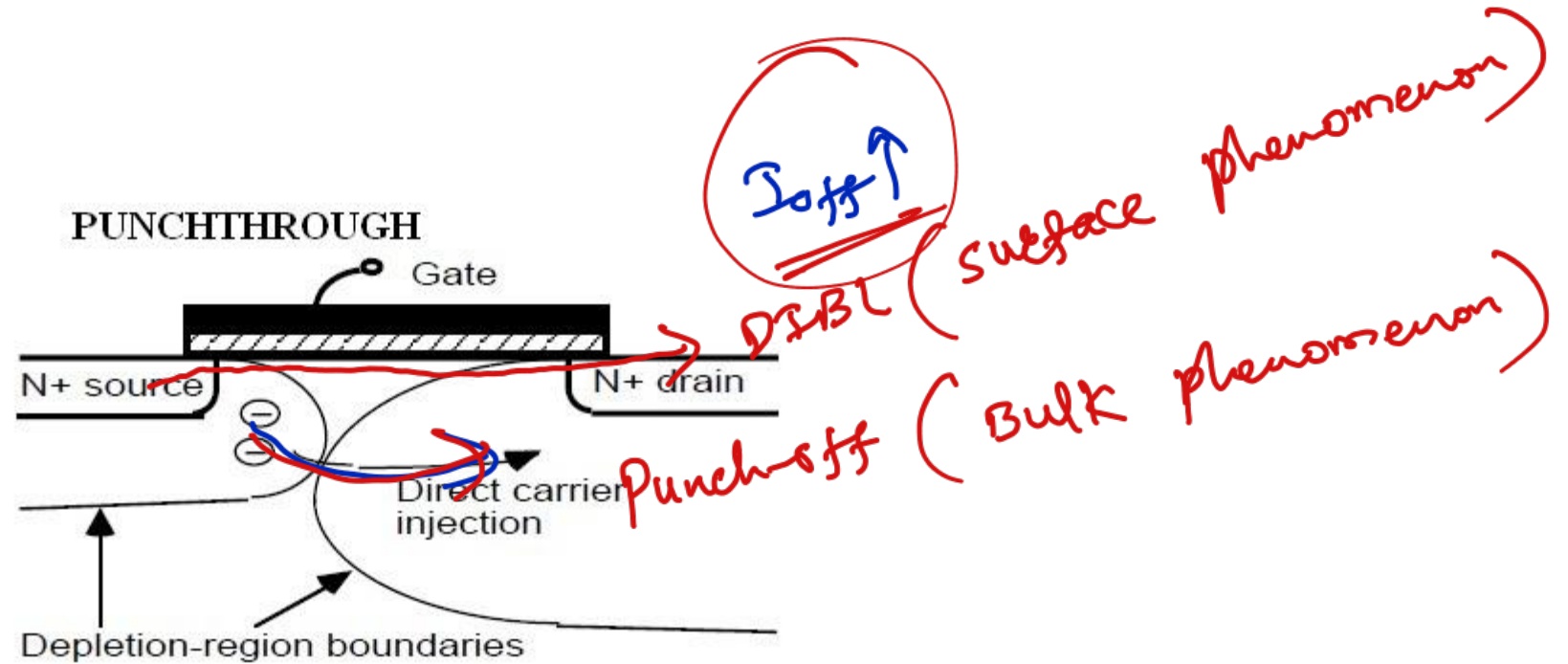
3. Drain Induced Barrier Lowering

(DIBL)



MOSFET Scaling – Short channel Effects

4. Punch Through



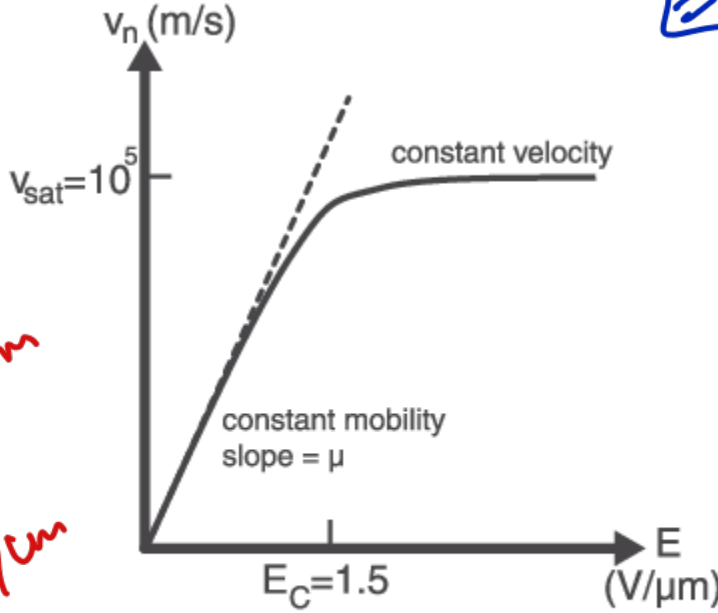
MOSFET Scaling – Short channel Effects

5. Velocity Saturation

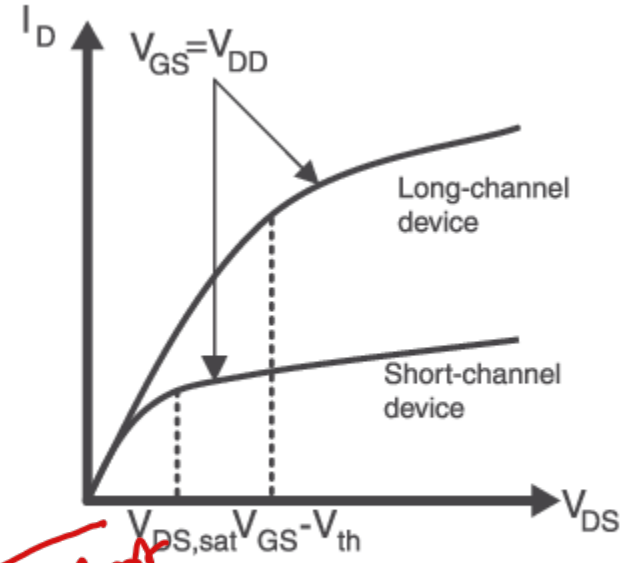
Handwritten notes:

$$E = \frac{1}{10nm} = 10^8 \text{ V/m}$$

$$\sim 10^6 \text{ V/cm}$$



(a)



(b)

Handwritten notes:

$v_{avg} \rightarrow v_{sat}$

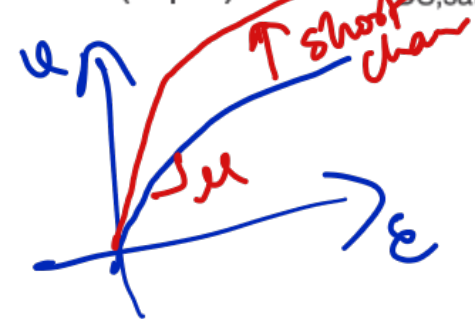
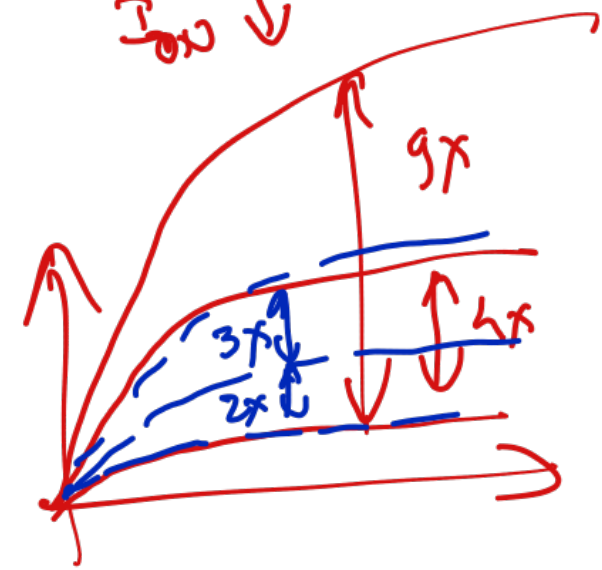
Ballistic transport

Handwritten notes:

$$I_{ON} \propto (V_{GS} - V_{TH})^2$$

$$I_{ON} \propto (V_{GS} - V_{TH}) V_{sat}$$

$v \downarrow$
 $I_{ON} \downarrow$



Handwritten notes:

$$v \sim \mu E$$

MOSFET Scaling – Short channel Effects

6. Gate induced drain leakage

"GIDL"

